

Inside the Head — what an AI agent's brain actually is

What is actually happening inside an AI agent's head?

In most diagrams, the answer is a label. A box that says "reasoning." A box that says "memory." Nothing under the boxes.

I drew the head differently. Every region traces to a kernel on disk. Every kernel has a SHA-256 hash that replays five times byte-identical.

PREFRONTAL — yuyay_v3, hash bacf5443. 430 SLOC. The wisdom gate. 13 axes evaluated AND, not averaged. moralGrounding and measurabilityHonesty must clear 0.95. Seven structural axes must each clear 0.90. Either the thought passes or it dies in the cortex.

FRONTAL — RIMAY. Proposer. Composes action candidates. Does not commit. Hands to parietal for simulation, then to heart for the gate.

TEMPORAL — yachay codex. Retrieval / RAG. Every fact carries a continuum_hash so the action chain and the evidence chain share a cryptographic spine.

PARIETAL — MUSQUY, 219 SLOC. K-candidate simulator. Generates counterfactual variants so the heart operates with comparison, not in a vacuum.

OCCIPITAL — NAWI. Tool surface (MCP). Tool calls do not bypass the heart.

QUANTUM MIND — qm, ea65ddc5. 163 SLOC. Honest scope: NOT a quantum-physical claim. Density-matrix formalism (real 4×4 matrices) on classical CPUs. Invariant lam_min ≥ 0.225 — the smallest eigenvalue of the state never collapses below that floor, or the candidate set has degenerated.

OVERWATCH — r0513, df4e9741. 146 SLOC. Read-only. Five invariants. Watches every cycle. Halt authority belongs to HUKLLA. Separation of powers, in code.

The brain does not write. This is the most important sentence about the head. The brain reads YAWAR by a single tether. RUWAY writes. SENTRA inspects inline. YAWAR records.

R0513 watches. Six organs. One direction.

Sovereign seal — HATUN crown. Identity-trace to a human principal. 10-tripwire egress. Byte-deterministic commit. 5× replay verified.

What this is NOT (doctrine requires saying it):

- Not a neuroscience model. Anatomical names are geography for legibility.
- Not a frontier-throughput claim. Substrate (Cerebras, SambaNova, vLLM) is the muscle.
- Not a quantum-physical claim. Classical hardware throughout.
- Not yet wired end-to-end. The 13-axis HEART is a sibling kernel; on-disk HATUN-RAID still calls a 9-axis YUYAY envelope. That migration is the next ratchet. I will not draw the upgrade as done until receipts say it is done.

If the diagram and the disk disagree, the disk wins. The diagram changes. The receipts do not lie.

Receipts available on request.

— Stephen P. Lutar · SZL Holdings · ORCID 0009-0001-0110-4173

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